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Assessing soil nutrient spatial variability in Mexico through digital soil mapping: Implications for sustainable agriculture

Thalita Fernanda Abbruzzini [a,b,*], Daniela Figueroa [a,b], Carlos Eduardo Arroyo Cruz [c], Viviana Marcela Varón-Ramírez [c,d], Mario Antonio Guevara Santamaria [b,c,*], Blanca Prado [a,b]

[a] Instituto de Geología, Departamento de Ciencias Ambientales y del Suelo, Universidad Nacional Autónoma de México, C.P. 04510 Ciudad de México, Mexico
[b] Programa Universitario de Estudios Interdisciplinarios del Suelo, PUEIS-UNAM, Mexico
[c] Instituto de Geociencias, Universidad Nacional Autónoma de México, C.P. 76230, Campus Juriquilla, Querétaro, Mexico
[d] Corporación Colombiana de Investigación Agropecuaria, AGROSAVIA, C.P. 250047 Sede Tibaitatá, Cundinamarca, Colombia

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ABSTRACT

The availability and spatial variability of soil nutrients strongly influence ecosystem services, food production, and sustainable land management, yet major information gaps persist at national and global scales. This study developed the first national digital maps of soil properties and nutrients for Mexico, contributing to the FAO's Global Soil Nutrient and Nutrient Budget Map (GSNmap) initiative. Using approximately 7000 harmonized soil observations at 0–30 cm depth and a comprehensive set of environmental covariates, we applied Quantile Regression Forest (QRF) models to predict soil texture (sand, silt, and clay contents), bulk density (BD), pH, cation exchange capacity (CEC), organic carbon (OC) and total nitrogen (N) contents, available phosphorus (P), and exchangeable potassium (K) and calcium (Ca) concentrations at 250 m resolution. Model performance was heterogeneous but overall robust at the national scale. Soil texture fractions, pH, CEC, OC, K, and Ca were predicted with moderate to strong accuracy ($R^2 \approx 0.50$–0.77). In contrast, N and P exhibited lower predictive accuracy ($R^2 \leq 0.26$), reflecting their strong sensitivity to localized management practices and land-use history, and suggesting that more intensive sampling strategies are required to capture their spatial variability. Independent hold-out validation confirmed the generalizability of model predictions across agricultural lands and grasslands. The resulting maps reveal pronounced regional contrasts. The Yucatán Peninsula is characterized by high CEC and OC but low P availability associated with karstic, carbonate-rich soils; northern regions exhibit low OC and N and increased vulnerability to structural degradation and salinization under arid, irrigation-dependent systems; and Central-South Mexico faces constraints related to soil acidity in predominantly rainfed agriculture. Differences among agricultural systems were also evident, with cultivated pastures showing higher OC and CEC than rainfed croplands, highlighting the influence of land use and management on soil functioning. By delivering spatially explicit predictions together with uncertainty estimates, this study provides a national baseline for diagnosing soil fertility constraints and supports region-specific soil management strategies. The results establish an operational and uncertainty-informed framework for soil nutrient mapping in Mexico, strengthening the linkage between national soil information systems and global initiatives to support sustainable soil governance and food security planning.

1. Introduction

Sustainable management of agricultural soils is fundamental for societies, underpinning food security, biodiversity, and water regulation. Soil fertility depends on macronutrients such as nitrogen (N), phosphorus (P), and potassium (K), along with micronutrients like iron, manganese, and zinc (FAO, 2022a). Their availability, together with soil physical and chemical properties, governs ecosystem processes by shaping carbon and water cycles, species diversity, and trophic interactions (Van Sundert et al., 2020). Soil fertility is also tightly linked to water dynamics, as soils regulate plant water supply and sustain broader ecosystem services.

Agricultural sustainability is increasingly threatened by climatic and socio-economic contexts that directly affect soil health (Figueroa et al.,

* Corresponding authors.
E-mail addresses: thalita@geologia.unam.mx (T.F. Abbruzzini), mguevara@geociencias.unam (M.A.G. Santamaria).

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2020; Streimikis and Baležentis, 2020). Rising temperatures and shifting rainfall patterns alter nutrient cycling (Corwin, 2021), while management decisions depend on both crop nutrient demand and farmers' financial capacity (Hou et al., 2020). The widespread use of irrigation, mechanized tillage, and chemical fertilizers increased global food production by 52% between 2000 and 2020, while reducing land demand by 18% (FAO, 2022b). Nonetheless, these advances entailed severe environmental costs, including soil degradation, water pollution, greenhouse gas emissions, and biodiversity loss (Montalvo-Romero et al., 2023).

Globally, nutrient overuse has disrupted N and P cycles, which are now among the most transgressed planetary boundaries (Richardson et al., 2023). These imbalances also affect the carbon cycle, influencing productivity, soil organic matter, and greenhouse gas emissions, thereby driving climate impacts, food insecurity, and biodiversity loss (Sutton et al., 2013; Vicca et al., 2018; Ortiz-García et al., 2022). In Mexico, these pressures are acute since ~64% of soils show some level of degradation and 12% are severely degraded (CONAFOR–UACh, 2013), largely due to erosion, salinization, and inadequate management (Santini et al., 2022). Fertilizers have improved yields but reduced natural fertility and increased reliance on external inputs (Tan et al., 2005; Galloway et al., 2008). Land-use change and deforestation have further contributed to desertification across nearly 60% of the territory (Granados-Sánchez et al., 2013). These trends underscore the urgency of robust, spatially explicit soil information to support sustainable management.

Climatic heterogeneity further complicates soil management. Northern Mexico has experienced significant warming and intensified rainfall patterns, while central and southern regions face declining precipitation (Cuervo-Robayo et al., 2020). These contrasting trends modify evapotranspiration, soil properties, and nutrient availability (Corwin, 2021). They also underpin two distinct agricultural models: intensive, commercial irrigated systems in the north, versus small-scale, rainfed and often subsistence farming in the south (Galeana-Pizaña et al., 2021). Despite their differences, both systems rely heavily on soil nutrient reserves. However, critical gaps in national soil data persist, underscoring the need for up-to-date, spatially continuous information to close yield gaps, promote sustainability, and inform policy (FAO, 2022a).

Although Mexico has a long history of territorial information (Ramírez-Jaramillo et al., 2023), soil-related data remain limited and fragmented, restricting their effective use in decision-making. Valuable resources exist—notably the INEGI's Edaphological Series I–III (INEGI, 2004, 2013, 2024)—which provide the most comprehensive soil databases in the country and have supported national inventories and soil carbon assessments (e.g., Paz-Pellat and Velázquez-Rodríguez, 2018; Ricker et al., 2020; Arroyo-Cruz et al., 2025). At the global scale, standardized products such as SoilGrids (Poggio et al., 2021) provide useful references in regions where national soil information is scarce or inaccessible. However, these datasets are largely based on compilations of point observations or generalized global models and are not designed to capture country-specific management practices or nutrient dynamics. To date, no spatially explicit, national-scale soil nutrient maps are available for Mexico.

This gap underscores the need for tailored Digital Soil Mapping (DSM) approaches that integrate harmonized datasets with local expertise. DSM is a computer-assisted branch of soil cartography. On DSM, soil hard data (i.e., expensive, time consuming) is coupled in the statistical space with proxies of soil forming factors to predict soil property or class values across an entire area of interest (McBratney et al., 2003; Hengl, 2003; Scull et al., 2003). Using remote sensing, digital terrain analysis and statistical learning (e.g., geostatistics, machine learning) DSM has become the most cost-effective approach to characterize the spatial variability of soil property values or classes across scales. Integration of DSM with local expertise is particularly appealing for soil nutrient modeling and mapping.

Soil nutrient availability depends on factors such as organic matter, texture, cation exchange capacity (CEC), and pH (Van Sundert et al., 2020). To capture this variability, the FAO's GSNmap initiative promotes DSM to harmonize data and generate spatially explicit nutrient information (FAO, 2022a). A key advantage of DSM lies in its ability not only to produce continuous maps but also to quantify prediction uncertainty, thereby improving the transparency and reliability of soil information for decision-making. Within this framework, Quantile Regression Forests (QRF; Meinshausen, 2006) are recommended in FAO guidelines for their ability to provide both predictive outputs and uncertainty estimates. QRF is a variant of Random Forests (RF)—a popular ensemble of regression trees based on bagging—(Breiman, 2001), which is a well known machine learning algorithm due its relatively high performance compared to other prediction algorithms on DSM (Vaysse and Lagacherie, 2017; Guevara et al., 2018). Consequently, the use of this algorithm represents a competitive statistical benchmark for wall-to-wall DSM mapping efforts in Mexico.

Accordingly, this study applies a DSM framework based on QRF to predict the spatial distribution and uncertainty of key soil properties and essential nutrients across agricultural and grassland soils in Mexico. By generating the first national-scale nutrient maps, we aim to (i) establish a harmonized and spatially explicit baseline for soil management, (ii) compare variability across agricultural systems and economic regions, and (iii) assess model performance and uncertainty. We hypothesize that QRF-based DSM can reliably capture nutrient variability across agro-climatic and governance gradients, producing distinctive spatial patterns that reflect management practices and regional contexts. By testing this, the study advances beyond previous inventories toward predictive, uncertainty-informed nutrient mapping, strengthening both Mexico's soil governance and the FAO's global GSNmap initiative.

2. Materials and methods

2.1. Study site

2.1.1. Geographic and biophysical context

This study was conducted at the national scale, encompassing all agricultural and grassland systems in Mexico (Fig. 1). The country spans latitudes 14°–32°N, where the Nearctic and Neotropical biogeographic regions converge, resulting in highly diverse topographic, climatic, and ecological conditions (Barreras et al., 2023). Mexico harbors 28 of the 32 soil groups recognized by the IUSS (2015), making it one of the most pedodiverse countries globally. Shallow and weakly developed soils such as Leptosols, Regosols, and Calcisols cover more than half of the territory, while more fertile soils such as Phaeozems, Luvisols, and Vertisols account for roughly one-third. This broad pedodiversity creates both challenges and opportunities for DSM, as nutrient availability is strongly conditioned by local soil–landscape interactions.

2.1.2. Agricultural systems, land use, and soil degradation in Mexico

Mexico's agricultural frontier covers ~51 million ha, with ~65% dedicated to crops and the remainder to livestock production (INEGI, 2021). Farming systems are classified into rainfed, irrigated, and moisture-dependent types, while grasslands are distinguished as cultivated or induced (INEGI, 2023) (Table 1, Fig. 1c). Maize dominates the landscape, occupying ~32% of agricultural land and underpinning national food security, but other cereals, sugarcane, fruits, and vegetables also play central roles in export-oriented production (SIAP–SADER, 2024). These systems exhibit large productivity gaps between irrigated and rainfed areas, reflecting contrasting access to inputs and infrastructure. Soil degradation is widespread, with erosion on sloping rainfed lands, salinization in irrigated valleys, and overgrazing in grasslands identified as key processes (Santini et al., 2022; Jurado-Guerra et al., 2021). Approximately two-thirds of national soils show some degree of degradation, and climate change is projected to further reduce yields of staple crops such as maize by 10–20% by 2050 (Gang et al., 2014; Murray-

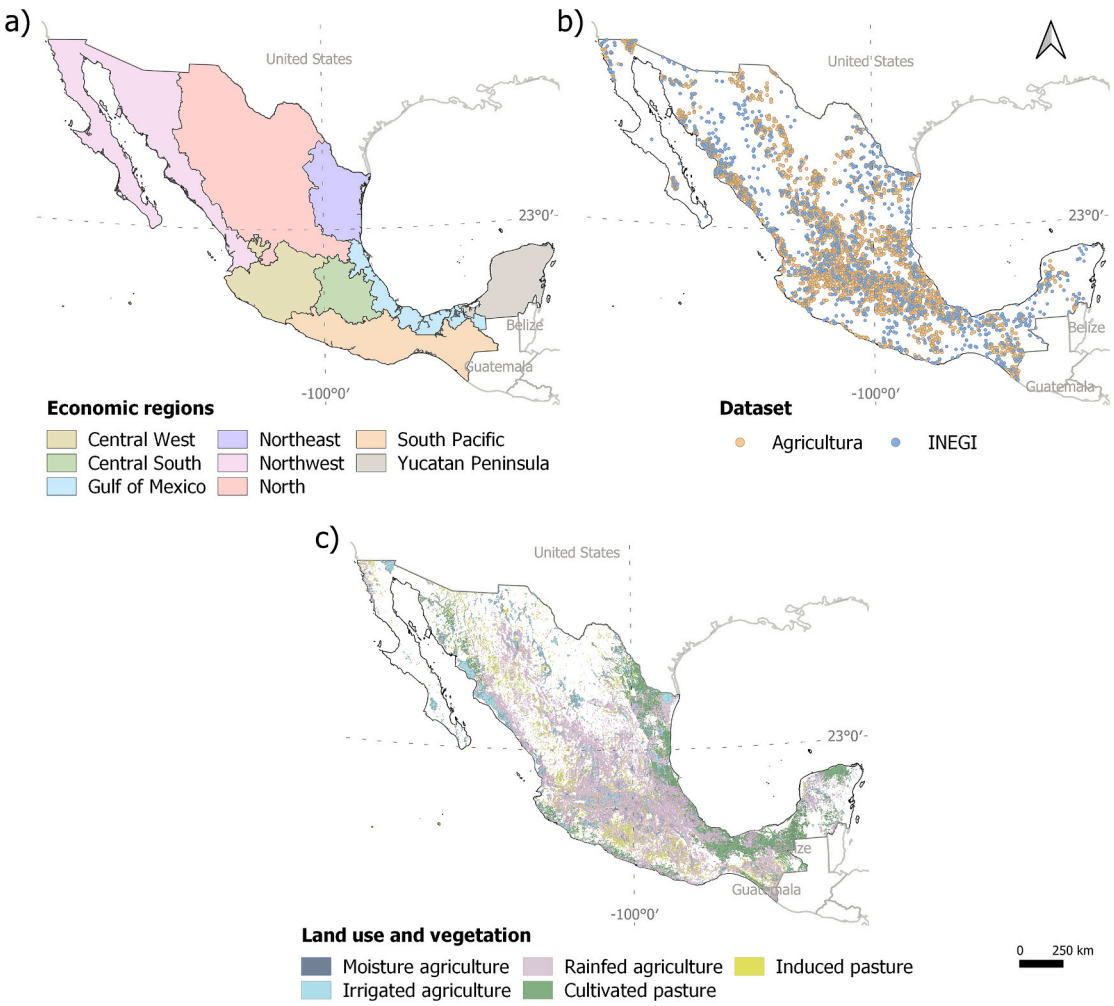


Fig. 1. a) Economic regions of Mexico. **Central West**: Jalisco, Colima, Michoacán, Guanajuato and Aguascalientes. **Central South**: Querétaro, Hidalgo, Estado de México, Ciudad de México, Morelos, Puebla and Tlaxcala. **Gulf of Mexico**: Veracruz and Tabasco. **Northeast:** Nuevo León and Tamaulipas. **Northwest:** Baja California, Baja California Sur, Sonora, Sinaloa and Nayarit. **North:** Chihuahua, Durango, Coahuila, San Luis Potosí and Zacatecas. **South Pacific:** Guerrero, Oaxaca and Chiapas. **Yucatán Peninsula:** Campeche, Yucatán and Quintana Roo. **(b)** Spatial distribution of soil observation datasets compiled from AGRICULTURA and INEGI. **(c)** Land use and vegetation types across Mexico used for soil analysis and mapping (INEGI Series V). All maps are displayed in WGS84 / EPSG:4326, with coordinate grid, scale bar, and north arrow.

Tortarolo et al., 2018). These pressures reinforce the need for robust, spatially explicit soil information to support sustainable management strategies.

2.1.3. *Regional and administrative framework*

Mexico's heterogeneity is expressed not only in its soils and climates but also in production practices, land tenure, and socio-environmental interactions (Ortiz-García et al., 2022). To account for this diversity, the country is commonly divided into eight economic regions, each grouping states with shared geographic and historical characteristics (Fig. 1a) (Comisión Nacional para el Conocimiento y Uso de la Biodiversidad (CONABIO), 2010). Unlike ecoregions, which seldom align with administrative boundaries (Voyloshnikova, 2020), economic regions reflect long-term socioeconomic dynamics and provide a practical unit for environmental management. Classical regional geography (Bassols Batalla, 1979; Bassols Batalla, 1983) already emphasized that agricultural areas cannot be analyzed in isolation but as part of broader economic structures that shape production, exchange, and consumption (Aguilar Sánchez, 2021). Building on this perspective, economic regionalization offers a suitable framework to assess soil properties and nutrient content in relation to agricultural land use, highlighting shared management practices across croplands and pastures. This approach also aligns with state-level governance structures, enhancing the potential of soil nutrient maps to inform decision-making and to design region-specific strategies for sustainable management (Ortiz-García et al., 2023).

2.2. *Digital soil mapping framework*

2.2.1. *Soil property datasets*

To calibrate and validate the DSM models, we first assembled harmonized soil observational datasets with georeferenced coordinates from the Ministry of Agriculture and Rural Development (AGRICULTURA) (Arroyo-Cruz et al., 2025) and the National Institute of Statistics and Geography (INEGI) (Fig. 1b). Available variables included soil texture (sand, silt, clay), bulk density (BD), pH, organic carbon (OC), cation exchange capacity (CEC), total nitrogen (N), available phosphorus (P), and exchangeable potassium (K) and calcium (Ca) for the 0–30 cm layer. Laboratory analyses followed the Official Mexican Norm NOM-021-RECNAT (SEMARNAT, 2002), which establishes standardized protocols for soil fertility and classification; these procedures were consistently applied across all samples (Table 2).

The number of observations varied among soil properties: most variables had ~7000 records, while P ($n = 3594$), BD ($n = 4461$) and N

Table 1
Main Types of Agricultural Land Use and Pastures in Mexico.

Type	Description
Rainfed Agriculture	Farming systems that rely solely on natural rainfall to supply the water required for crop growth and vegetative development. This practice is common in areas where irrigation infrastructure is unavailable or unnecessary, and crop productivity depends heavily on seasonal and climatic rainfall patterns.
Irrigated Agriculture	Farming systems where water is supplied to crops through controlled methods from external sources, such as wells, dams, rivers, reservoirs, or irrigation systems, to supplement natural rainfall. This practice is implemented to ensure consistent crop growth, especially in regions with insufficient or irregular precipitation.
Moisture-Dependent Agriculture	This type of agriculture is ancient and usually depends on the traditional knowledge of rural communities in Mexico. Regardless of the rainfall cycle, this type of agriculture takes advantage of soil moisture and retains it even during dry periods—for example, in flood-prone areas such as reservoir beds. The typical example of this type of agriculture are the *chinampas*; an artificial agricultural system constructed in zones where the water is the main surrounding natural resource.
Cultivated Pasture	Grassland systems where non-native grass species have been deliberately introduced into a region to enhance productivity, improve forage quality, or support soil conservation efforts. These systems are actively managed and maintained through practices such as fertilization, irrigation, or grazing control to optimize their ecological or economic benefits.
Induced Pasture	Grasslands created or maintained through human intervention, often as a result of land-use changes like deforestation, logging, or intentional burning. These pastures may emerge naturally as a transitional phase in the ecological succession of plant communities but are often artificially sustained or perpetuated due to repeated and intensive disturbances such as livestock grazing, soil compaction, or fire management practices.

Table 2
Soil properties, units, and laboratory determination methods.

Soil property	Unit	Calculation method
Soil texture (sand, silt, clay)	%	Bouyoucos hydrometer
Bulk density (BD)	g cm^{-3}	Cylinder method
pH	–	2:1 soil-water ratio
Cation exchange capacity (CEC)	$cmol_c$ kg^{-1}	Ammonium acetate AS-12
Organic Carbon (OC)	%	Walkley and Black multiplied by the Van Bemmelen factor (1.724) (Sparks, 1996)
Total nitrogen (N)	%	Wet digestion and Kjeldahl distillation
Available phosphorus (P)	mg kg^{-1}	Olsen or Bray, depending on the pH estimation
Exchangeable potassium (K)	$cmol_c$ kg^{-1}	Ammonium acetate pH 7.0, Flame photometry or Atomic absorption
Exchangeable Calcium (Ca)	$cmol_c$ kg^{-1}	

(n = 3900) were less represented, as these were not consistently reported across the INEGI series. To ensure consistency with remote sensing covariates, we retained only samples collected between 2000 and 2017 (Fig. 1b). For INEGI's profile data, we applied a mass-preserving spline function (Malone et al., 2009) to interpolate soil properties across depth intervals, following GlobalSoilMap.Net project specifications. The first three standardized depths were extracted for model training. Finally, quality control included data cleaning and the removal of outliers following the threshold criteria of Schuenemeyer and Drew (2011). For N specifically, values exceeding 1% were excluded to avoid unrealistic extremes.

Then, we applied a DSM framework to predict soil nutrients and associated properties for the 0–30 cm depth at a 250 m resolution across Mexico's agricultural lands. The modeling approach was based on QRF (Meinshausen, 2006; Vaysse and Lagacherie, 2017). This algorithm was recommended by the FAO's Global Soil Partnership for the GSNmap initiative (FAO, 2022a), as it is particularly suited to digital soil mapping. QRF does not assume a predefined functional form between covariates and soil properties, and it provides full conditional distributions (e.g., 5th and 95th percentiles) in addition to mean estimates. This allows the derivation of prediction intervals and explicit assessment of uncertainty, thereby increasing the transparency and robustness of DSM products.

2.2.2. Covariate selection and data harmonization

To develop digital maps of soil properties and nutrients, we compiled a comprehensive set of environmental covariates aligned with the SCORPAN framework (Table S1). These covariates included: **S** (soil): profile top and bottom depths; **C** (climate): 19 bioclimatic variables (BIO1-BIO19) capturing temperature and precipitation regimes following the methodology of Flick and Hijmans (2023) and generated for Mexico using the high-resolution climate datasets of Cuervo-Robayo et al. (2020); **O** (organisms/land cover): Normalized Difference Vegetation Index (NDVI) metrics (mean, maximum, coefficient of variation) and agricultural land-use type; **R** (relief, topography): altitude, slope, curvature indices, catchment properties, topographic wetness index (TWI), multi-resolution ridge top flatness (MRRTF), and multi-resolution valley bottom flatness (MRVBF); **A** (age): year of observation (used as an operational proxy of temporal heterogeneity rather than pedogenetic soil age); and **N** (spatial position): latitude and longitude. All covariates were harmonized to a 250 × 250 m spatial resolution: continuous variables were resampled using bilinear interpolation, while categorical variables were resampled with a nearest-neighbor approach. Data layers were aligned to a unified coordinate reference system, and each soil observation was paired with the corresponding covariate values to generate the regression matrix used for model calibration.

2.2.3. Model calibration and prediction

We first applied a Recursive Feature Elimination (RFE) strategy with repeated cross-validation to optimize model parameters and identify the most informative predictors. In each iteration, RFE evaluates the effect of removing individual variables on prediction error, retaining those whose absence most increases model error as the most informative predictors. The predictive framework was based on the RF algorithm (Breiman, 2001), widely used in DSM for its robustness, non-parametric nature, and ability to capture non-linear relationships between soil properties and environmental covariates (Vaysse and Lagacherie, 2017). Comparative reviews highlight that RF consistently performs better or at least as well as alternative algorithms in DSM applications, providing higher predictive accuracy and stability across different soil properties and covariate sets (Lamichhane et al., 2019; Pouladi et al., 2023). RF builds an ensemble of regression trees using bootstrap samples and random subsets of predictors (Breiman, 1996); predictions are obtained as the ensemble average. Hyperparameters tuned included the number of trees required for error stabilization and the number of predictors randomly sampled at each node split.

We implemented the QRF variant (Meinshausen, 2006), which extends RF to estimate full conditional distributions, thereby providing both central tendency (mean/median) and uncertainty (prediction intervals) for each pixel. Final models were trained using the most informative covariates retained by RFE and the parameter set minimizing prediction error after cross-validation. This process generated spatially explicit maps of mean values of predicted soil properties and nutrients, along with corresponding uncertainty surfaces, across agricultural lands and grasslands of Mexico.

Model performance was evaluated using two complementary validation strategies: First, a 10-fold cross-validation was implemented within the training dataset (75% of the observations) to assess internal predictive stability. Second, an external hold-out validation was performed, in which the remaining 25% of the observations—randomly withheld prior to model fitting—were used exclusively for independent evaluation of model predictions. All analyses were performed in R (R

Core Team, 2024) using the packages *quantregForest* (Meinshausen, 2006) for QRF implementation, *caret* (Kuhn, 2008) for model training and validation, and *raster* (Hijmans, 2023) for spatial data processing.

2.2.4. Accuracy indicators and variable importance

To quantify model accuracy, we used a comprehensive set of performance indicators commonly applied in DSM (Wadoux et al., 2022). Model performance was primarily evaluated using the coefficient of determination (R^2), which describes the proportion of variance explained, and the root mean squared error (RMSE), which reflects the overall magnitude of prediction errors. Agreement between observed and predicted values was evaluated using the Pearson correlation coefficient (r), the concordance correlation coefficient (CCC), the coefficient of efficiency (COE), and the index of agreement (IOA; Willmott, 1981). In addition, FAC2 was reported to summarize the proportion of predictions falling within a factor of two of the observed values. Together, these indicators provide a robust assessment of model accuracy, bias, and agreement between observations and predictions.

Variable importance was quantified via permutation importance, expressed as the percentage increase in mean squared error (%IncMSE), defined as the rise in prediction error when a predictor is randomly permuted, relative to the baseline out-of-bag (OOB) error (Breiman, 2001; Wei et al., 2015). Larger %IncMSE indicates greater predictive contribution. Final predictions for each soil property and nutrient were generated as continuous raster maps at 250 m resolution across all agricultural lands and grasslands in Mexico.

2.3. Statistical comparison across agricultural systems and economic regions

We selected polygons corresponding to the three types of agriculture and the two types of pasture (Fig. 1c) reported in the land use and vegetation cover layer for Mexico (Series V) (INEGI, 2015). Although INEGI Series VII provides the most recent national reference (INEGI, 2021) and was used to delineate the overall extent of agricultural and pasture systems, Series V was specifically employed for comparative analyses among systems due to its closer temporal alignment with the soil sampling dates used for model calibration.

For each soil property and nutrient (sand, silt, clay, BD, pH, OC, CEC, N, P, K, and Ca), zonal statistics were calculated, including the mean and standard deviation as well as minimum and maximum values. Geometry corrections were subsequently applied to the resulting layers. Using an intersection algorithm, we then overlaid these results with the economic regions of Mexico to calculate areas in hectares and report soil property dynamics across agricultural and pasture systems. Spatial analyses were conducted using the open-source software QGIS 3.36.

To complement the descriptive statistics, differences among agricultural systems and economic regions were evaluated using linear models with factorial designs including main effects and interactions. Analysis of variance (ANOVA) was used to test the significance of these effects. Subsequently, estimated marginal means (EMMs) with their standard errors (SEs) were derived, and Tukey-adjusted pairwise comparisons were performed to identify significant contrasts between groups. Sample sizes were large and consistent across soil variables for all regional, state, and system-level comparisons, supporting robust inference. Results were summarized using significance groupings, highlighting regional and management-based differences in soil nutrient availability.

3. Results

3.1. Accuracy assessment and covariate influence

The accuracy assessment of the QRF models revealed heterogeneous but overall robust predictive performance across soil properties and nutrients (Table 3; Fig. S1), with variable importance analyses (% IncMSE; Figs. S2–S4) providing complementary insight into the dominant environmental controls. Soil texture variables showed the most consistent and reliable predictions at the national scale, with strong agreement between observed and predicted values ($r = 0.71$–0.78; $R^2 = 0.50$–0.60; IOA = 0.69–0.73; FAC2 $\geq$ 0.91; Table 3). Observed–predicted relationships exhibited stable linear patterns across the full range of values (Fig. S1). Variable importance analyses (Fig. S2) indicate that these patterns are primarily driven by broad-scale climatic gradients, such as temperature and precipitation seasonality, in combination with physiographic attributes, such as elevation and terrain-derived metrics.

Soil pH and BD also showed strong to moderate predictive performance (pH: $r = 0.85$; $R^2 = 0.73$; IOA = 0.81; BD: $r = 0.63$; $R^2 = 0.40$; IOA = 0.65; Table 3), with low dispersion and narrow error ranges for BD (RMSE = 0.13 g cm^{-3}; Fig. S1). For these properties, variable importance rankings (Fig. S3) highlight the dominant role of precipitation-related bioclimatic variables capturing both moisture availability and seasonal variability, complemented by vegetation proxies derived from NDVI, reflecting strong control by climate and land cover at broad spatial scales. Soil CEC and OC were predicted with moderate accuracy (CEC: $r = 0.75$; $R^2 = 0.56$; IOA = 0.73; OC: $r = 0.72$; $R^2 = 0.52$; IOA = 0.70; Table 3), with increasing dispersion at higher values (Fig. S1). Variable importance analyses indicate that, for both properties, terrain attributes related to landscape position and hydrological redistribution (e.g., MRVBF and catchment indices), together with climatic controls, play an important role in shaping the spatial structure of model predictions.

Macronutrients exhibited contrasting predictive behavior. Total N showed lower to moderate predictive performance ($r = 0.51$; $R^2 = 0.26$; IOA = 0.67; CCC = 0.42; Table 3), with reduced sensitivity at higher concentrations (Fig. S1). Variable importance patterns for total N (Fig. S4) indicate a distributed control involving hydrological landscape position (e.g., MRVBF and TWI), vegetation productivity proxies (NDVI), and precipitation-related bioclimatic variables, particularly those

Table 3
Cross-validated accuracy metrics for predicted soil properties in agricultural lands and grasslands of Mexico.

Soil property	Unit	n	FAC2	RMSE	r	R^2	CCC	P	COE	IOA
Sand	%	48,285	0.914	13.085	0.769	0.592	0.744	0.000	0.456	0.728
Silt	%	48,303	0.939	7.769	0.709	0.503	0.665	0.000	0.380	0.690
Clay	%	48,303	0.921	10.974	0.776	0.602	0.747	0.000	0.465	0.733
BD	g cm^{-3}	29,790	1.000	0.135	0.634	0.402	0.577	0.000	0.294	0.647
pH	Unitless	48,843	1.000	0.658	0.854	0.730	0.847	0.000	0.613	0.807
CEC	$cmol_c$ kg^{-1}	47,169	0.872	9.135	0.751	0.564	0.716	0.000	0.455	0.727
OC	%	46,143	0.884	0.569	0.719	0.518	0.671	0.000	0.396	0.698
N	%	26,325	0.859	0.085	0.512	0.262	0.419	0.000	0.345	0.672
P	mg kg^{-1}	24,282	0.563	13.888	0.438	0.192	0.319	0.000	0.181	0.591
K	$cmol_c$ kg^{-1}	46,386	0.836	0.476	0.713	0.509	0.666	0.000	0.412	0.706
Ca	$cmol_c$ kg^{-1}	47,844	0.798	9.562	0.835	0.698	0.820	0.000	0.609	0.805

FAC2: fraction of predictions within a factor of two of observations; RMSE: root mean square error; r: Pearson correlation coefficient; R^2: coefficient of determination; CCC: concordance correlation coefficient (Lin, 1989); P: p-value associated with r; COE: coefficient of efficiency; IOA: index of agreement (Willmott, 1981).

characterizing dry-season moisture availability. Available P displayed the lowest predictive performance (r = 0.44; R^2 = 0.19; IOA = 0.59; CCC = 0.32; FAC2 = 0.56; Table 3), with pronounced saturation effects at higher concentrations (Fig. S1) and weak, fragmented importance patterns dominated by climatic variables (Fig. S4), underscoring the limited capacity of environmental covariates to capture fine-scale, management-driven variability, compounded by the reduced sample size available for P relative to other soil properties. Exchangeable K and Ca showed moderate to high predictive performance (K: r = 0.71; R^2 = 0.51; IOA = 0.71; Ca: r = 0.84; R^2 = 0.70; IOA = 0.81; Table 3). Their spatial predictions are supported by coherent associations with climatic variables and landscape position metrics (Fig. S4), consistent with their more structured spatial distributions.

3.2. External (independent hold-out) validation and model transferability

Model performance was further evaluated using an independent hold-out dataset to assess external validation and model generalizability across agricultural lands and grasslands of Mexico (Table 4; Fig. S5). Accuracy metrics obtained from this independent evaluation were largely comparable to those derived from cross-validation (Table 3; Fig. S1), indicating that the QRF models successfully reproduce spatial patterns beyond the calibration dataset and the models are not affected by overfitting.

Soil texture fractions maintained strong predictive performance under external validation (R^2 = 0.47–0.63), with RMSE values similar to those observed during model calibration. Soil pH also showed consistently high performance (R^2 = 0.77; RMSE = 0.61), closely matching cross-validation results. BD, CEC, OC, and exchangeable Ca exhibited stable performance with only minor changes in accuracy metrics, reflected by small variations in RMSE, R^2, CCC, and IOA between cross-validation and external validation (Tables 3 and 4). As observed during cross-validation, total N and available P showed lower external validation performance, with reduced R^2 and IOA values relative to other soil properties. These patterns are consistent with their weaker performance during model calibration and reflect the combined effects of reduced sample size and strong control by fine-scale, management-driven processes not explicitly represented in the environmental covariate set. Nevertheless, predictions for N and P retained positive coefficients of efficiency and captured broad spatial gradients relevant for regional-scale assessments.

3.3. Comparisons and soil attribute analysis

Spatial predictions of soil properties and nutrients are shown together with their associated uncertainties. These maps provide a national overview of soil variability and uncertainty (see Figs. 2–4 and Figs. S6–S8), forming the basis for comparisons at regional, state, and land-use levels (Tables 5, 6 and 7, respectively). Unless otherwise stated, values reported in parentheses correspond to regional mean values.

3.3.1. Regional variability of soil properties in agricultural and grassland areas of Mexico

The analysis of variance (ANOVA) confirmed strong and highly significant effects of region on all soil variables ($p < 0.001$), indicating substantial regional variability across agricultural lands and grasslands (Table 5). Post-hoc comparisons (Tukey-adjusted EMMs) revealed consistent differences among regions for texture, pH, CEC, OC, P, K and Ca, while BD and N also differed significantly between regions but exhibited comparatively narrower numerical ranges and more moderate regional gradients. This suggests that, although BD and N respond to broad-scale edaphic and climatic controls, their spatial variability is more strongly modulated by local management and micro-scale processes. The large and relatively balanced sample sizes across regions ($n \geq 2533$; Table 5) provide robust support for these contrasts.

Textural patterns display clear regional gradients that reflect differences in parent material and depositional environments (Table 5; Fig. 2). Soils in the Yucatán Peninsula are markedly clay-rich (49.25% clay on average; 32.14% sand; 17.54% silt), contrasting with the coarser textures of the Northwest, South Pacific, North and Central South regions, where sand contents typically exceed 44% (e.g., Northwest: 48.77% sand; South Pacific: 46.51%; North: 45.53%; Central South: 44.07%). The Gulf of Mexico and Northeast regions show the highest silt contents (≈28%), while the Yucatán Peninsula has the lowest silt fraction, consistent with its dominance of highly weathered, fine-textured materials. These results highlight the strong imprint of regional geology and geomorphology on the distribution of soil texture across Mexico.

BD shows moderate regional variation (Table 5; Fig. 3), with slightly higher values in the Northwest (1.42 g cm^{-3}), North (1.35 g cm^{-3}) and Northeast (1.32 g cm^{-3}), and lower values in the Yucatán Peninsula (1.19 g cm^{-3}) and Gulf of Mexico (1.25 g cm^{-3}). These contrasts are consistent with the combined influence of texture, organic C content and management practices (e.g., tillage intensity and compaction). However, the relatively narrow BD range at the national scale suggests broadly comparable soil physical conditions within the main agricultural and pasture areas.

Soil pH exhibits a clear separation between more acidic and more alkaline regions (Table 5; Fig. 3). Central West, Central South, Gulf of Mexico and South Pacific are characterized by near-neutral to slightly acidic conditions (pH 6.32–6.69), reflecting more weathered parent materials and higher leaching. In contrast, the North, Northwest and Yucatán Peninsula display neutral to moderately alkaline conditions (pH 7.21–7.31), while the Northeast reaches clearly alkaline values (pH 8.19), likely associated with the presence of Ca and Mg carbonates and lower leaching. This pattern is consistent with the known distribution of calcareous soils in northern Mexico and the karstic landscapes of the Yucatán Peninsula.

CEC also shows marked regional differences (Table 5). The Yucatán Peninsula stands out with the highest CEC values (37.03 $cmol_c\ kg^{-1}$), well above the national mean (23.69 $cmol_c\ kg^{-1}$), indicating a high capacity to retain cationic nutrients. The Gulf of Mexico region also

Table 4
External hold-out validated accuracy metrics for predicted soil properties in agricultural lands and grasslands of Mexico.

Soil property	Unit	n	FAC2	RMSE	r	R^2	CCC	P	COE	IOA
Sand	%	1786	0.898	13.265	0.780	0.608	0.705	0.000	0.402	0.701
Silt	%	1788	0.930	8.142	0.685	0.470	0.610	0.000	0.320	0.660
Clay	%	1788	0.907	10.794	0.795	0.632	0.725	0.000	0.420	0.710
BD	g cm^{-3}	1114	1.000	0.130	0.674	0.454	0.600	0.000	0.269	0.635
pH	Unitless	1807	1.000	0.608	0.875	0.765	0.855	0.000	0.602	0.801
CEC	$cmol_c\ kg^{-1}$	1746	0.867	8.799	0.772	0.596	0.698	0.000	0.423	0.711
OC	%	1707	0.876	0.553	0.757	0.573	0.671	0.000	0.387	0.694
N	%	975	0.826	0.075	0.499	0.249	0.411	0.000	0.151	0.576
P	mg kg^{-1}	896	0.574	13.143	0.475	0.226	0.333	0.000	0.164	0.582
K	$cmol_c\ kg^{-1}$	1716	0.808	0.462	0.745	0.555	0.669	0.000	0.399	0.699
Ca	$cmol_c\ kg^{-1}$	1769	0.742	9.890	0.830	0.689	0.788	0.000	0.534	0.767

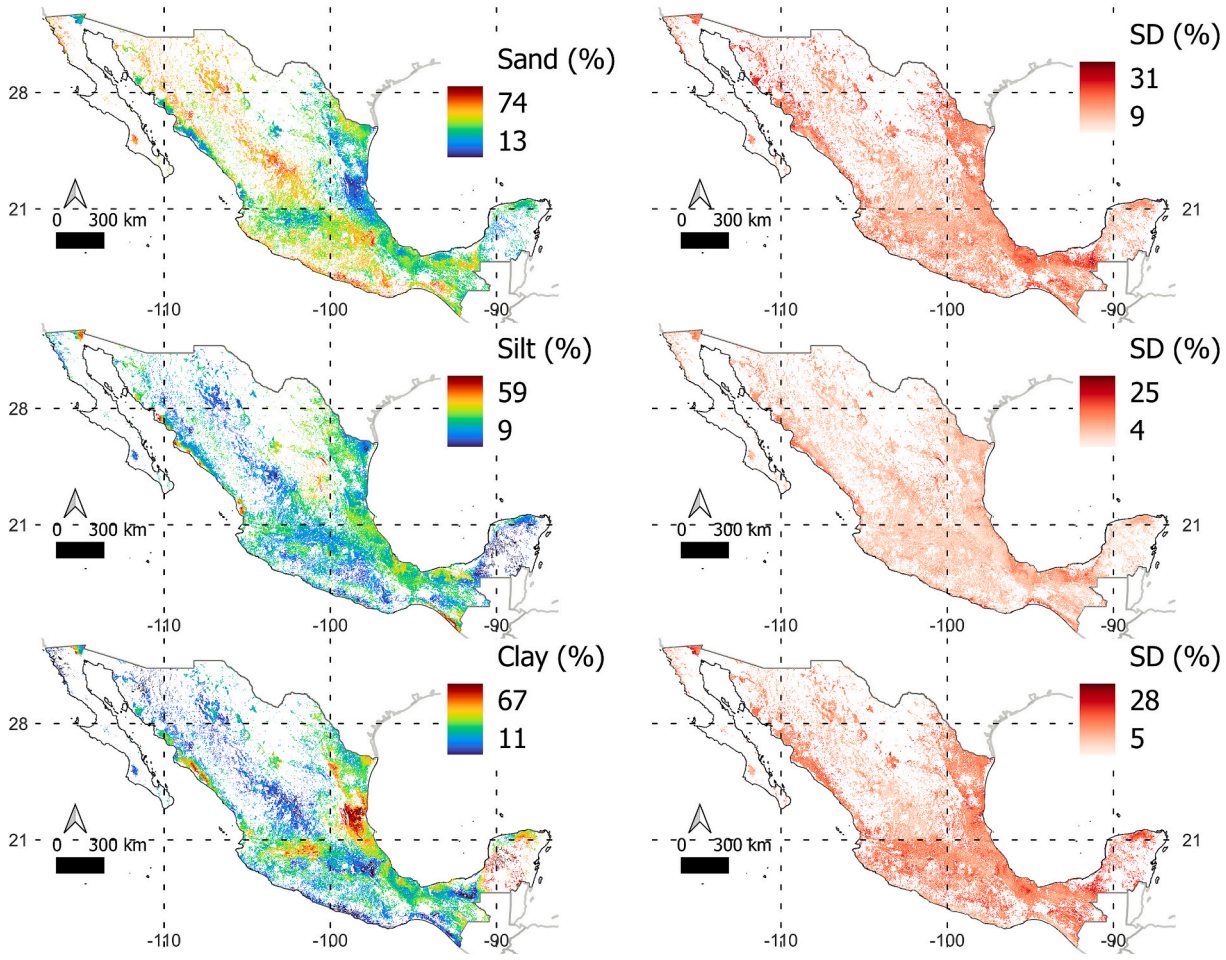


Fig. 2. Predicted mean and standard deviation of soil sand, silt, and clay contents across the economic regions of Mexico. Maps include coordinate grid, north arrow, and scale bar for spatial orientation (CRS: WGS84 / EPSG:4326).

exhibits relatively high CEC (27.51 $cmol_c\ kg^{-1}$), whereas northern regions such as Northwest, North and Northeast show lower values (≈21–22 $cmol_c\ kg^{-1}$). These contrasts reflect the combined effects of clay mineralogy, organic matter content and the presence of secondary carbonates.

The spatial patterns of soil OC, total N and exchangeable K broadly mirror those of CEC and texture (Table 5; Fig. 4). The Yucatán Peninsula shows the highest OC (2.13%) and N (0.22%) contents, followed by the Gulf of Mexico and South Pacific regions (OC ≈ 1.9%; N ≈ 0.16–0.17%), whereas the Northwest displays the lowest OC values (1.12%). Exchangeable K is likewise highest in Yucatán Peninsula (1.22 $cmol_c\ kg^{-1}$) and North (1.10 $cmol_c\ kg^{-1}$), and lowest in the Gulf of Mexico (0.66 $cmol_c\ kg^{-1}$) and South Pacific (0.51 $cmol_c\ kg^{-1}$). Available P exhibits a contrasting regional pattern: the North region shows the highest P levels (20.08 mg kg^{-1}), with relatively high values also in Central West, Central South, Northwest and South Pacific (≈17–18 mg kg^{-1}), while the Gulf of Mexico and especially the Yucatán Peninsula display markedly lower P contents (12.74 and 7.99 mg kg^{-1}, respectively).

Exchangeable Ca further emphasizes the edaphic contrasts between regions (Table 5; Fig. 4). The Yucatán Peninsula and Northeast show the highest exchangeable Ca contents (35.41 and 32.69 $cmol_c\ kg^{-1}$, respectively), consistent with their calcareous parent materials and alkaline pH, while Central West, Central South and South Pacific have substantially lower Ca levels (≈14–16 $cmol_c\ kg^{-1}$). Taken together, these patterns indicate that the Yucatán Peninsula concentrates a large share of Mexico's soil fertility in terms of OC, N, CEC, K and Ca, but with relatively low available P, while occupying only about 4–5% of the mapped agricultural and grassland area (2.28 Mha out of 51.82 Mha; Table 5). In contrast, several northern and central regions (e.g., North and South Pacific) cover a much larger fraction of the agricultural area yet exhibit coarser textures, lower OC and CEC, and more variable nutrient status.

3.3.2. *State-level variations in soil properties across Mexican agricultural and grassland areas*

Building on the regional analysis, the state-level assessment revealed substantial variability in soil properties across Mexican agricultural and grassland areas (Table 6). ANOVA detected highly significant differences among states for most variables ($p < 0.001$), and Tukey-adjusted post-hoc comparisons confirmed clear contrasts for soil texture, pH, CEC, OC, N, available P, exchangeable K and Ca. In contrast, BD did not show statistically significant differences among states, indicating relatively homogeneous values at this spatial scale aggregation level.

Textural composition varies markedly among states, although most exhibit intermediate sand contents, typically ranging between ~37% and ~ 50% (Table 6). States in northern and north-central Mexico tend to show higher sand proportions (e.g., Baja California, Sonora, Tlaxcala, Zacatecas and Chihuahua), whereas lower sand contents are observed in southeastern states such as Tabasco, Campeche and Yucatán. Silt contents generally range from ~22% to ~28%, with higher values in Tabasco and Veracruz and lower values in the Yucatán Peninsula states. Clay content spans a broader range (~23–51%), with the highest values concentrated in Campeche, Quintana Roo and Yucatán, reflecting strong state-level contrasts in parent material and pedogenic development.

BD shows a relatively narrow range across states (~1.11–1.52 g cm^{-3}; Table 6), with lower mean values in Yucatán and higher values in

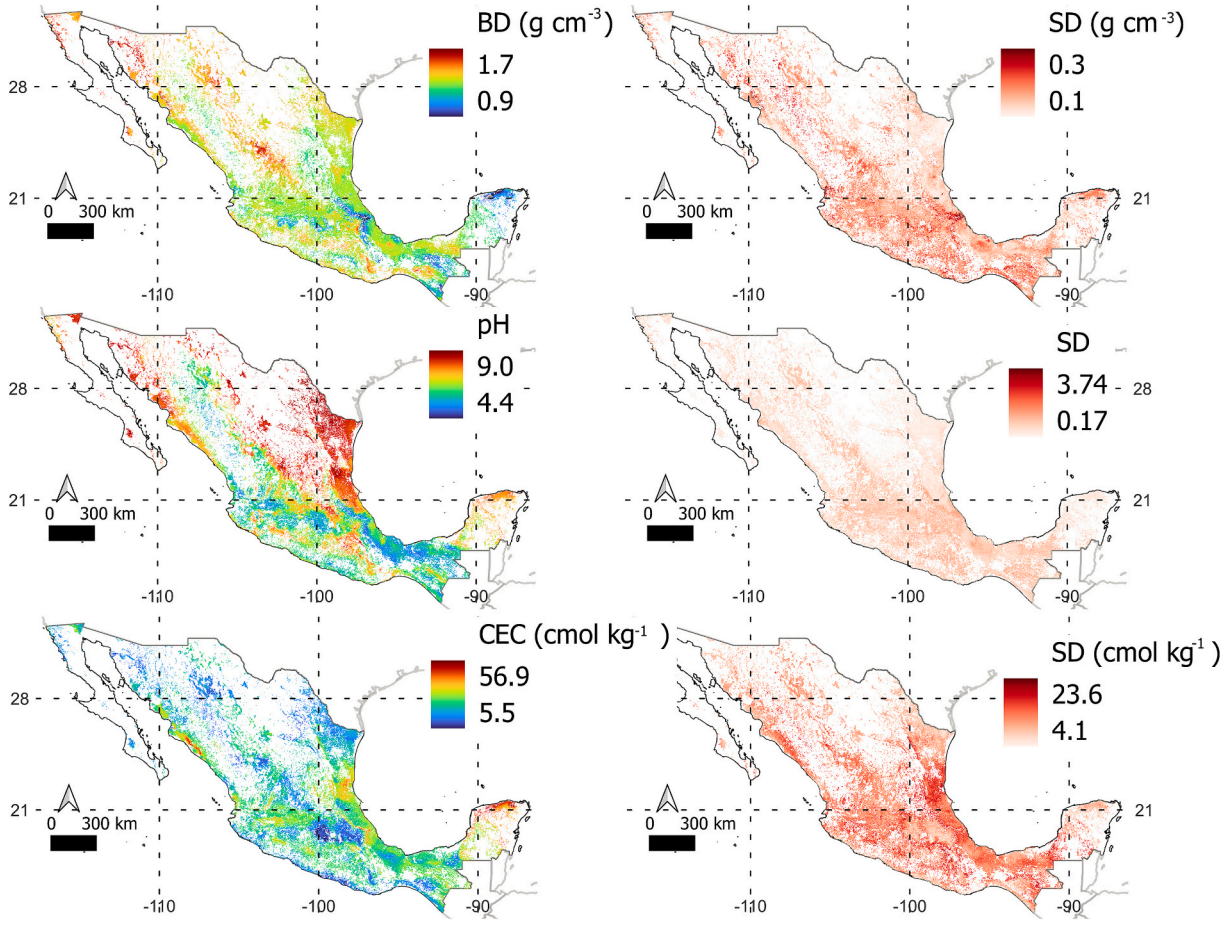


Fig. 3. Predicted mean and standard deviation of bulk density (BD), pH, and cation exchange capacity (CEC) across the economic regions of Mexico. Maps include coordinate grid, north arrow, and scale bar for spatial orientation (CRS: WGS84 / EPSG:4326).

Baja California, Baja California Sur and Sonora. However, these differences are not statistically significant, reinforcing the limited expression of BD variability at the state scale compared with other soil properties.

Soil pH exhibits clear state-level gradients, ranging from slightly acidic to moderately alkaline conditions (Table 6). Central and southern states such as Mexico City, State of Mexico, Chiapas and Nayarit show slightly acidic soils (pH ≈ 6.0–6.4), whereas northern states including Sonora, Baja California, Coahuila and Tamaulipas display neutral to alkaline conditions (pH ≥ 7.5). Within-state variability in pH is generally low, suggesting relatively consistent chemical environments at this spatial aggregation level.

CEC shows pronounced differences among states, with most values ranging between ~18 and 26 $cmol_c\ kg^{-1}$, but substantially lower values in some central highland states (e.g., Tlaxcala and the State of Mexico) and markedly higher levels in the Yucatán Peninsula. Yucatán (41.7 $cmol_c\ kg^{-1}$), Quintana Roo (37.9 $cmol_c\ kg^{-1}$) and Campeche (34.2 $cmol_c\ kg^{-1}$) stand out well above the national range, indicating a strong capacity for cation retention at the state level (Table 6).

Soil OC and total N also display strong state-level contrasts. OC contents range from ~0.6–1.0% in arid northern states such as Baja California, Baja California Sur and Sonora to >2.2% in Yucatán and Quintana Roo. Total N follows a similar pattern, with the highest values in Yucatán (0.35%) and Quintana Roo (0.28%), and substantially lower concentrations (<0.12%) in several northern states (Table 6).

Available P exhibits an almost inverse spatial pattern relative to OC and CEC. Several central and northern states, including Tlaxcala, Mexico City and Chihuahua, show relatively high P levels (~20–22 mg kg^{-1}), while the highest mean value is observed in Coahuila (26.36 mg kg^{-1}). In contrast, Yucatán and Quintana Roo exhibit consistently low P availability (<9 mg kg^{-1}). Exchangeable K varies significantly among states, from low values in Tabasco and Guerrero (~0.45–0.50 $cmol_c\ kg^{-1}$) to high levels in Yucatán and San Luis Potosí (>1.2 $cmol_c\ kg^{-1}$). Exchangeable Ca further emphasizes state-level edaphic contrasts, with particularly high concentrations in states underlain by calcareous or carbonate-rich parent materials, including Yucatán, Quintana Roo, Tamaulipas and Nuevo León (Table 6).

Overall, the state-level analysis reveals that much of the regional variability described in the previous section is preserved—and in some cases amplified—when examined at finer spatial resolution. States within the Yucatán Peninsula consistently concentrate the highest values of clay content, CEC, OC, N, K and Ca, while simultaneously exhibiting low available P, highlighting a distinct nutrient imbalance at this scale. In contrast, many northern and north-central states combine coarser textures, lower organic matter contents and more variable nutrient levels. The lack of statistically significant differences in BD across states, despite strong contrasts in texture and OC contents, suggests that this property is less sensitive to state-level aggregation and likely reflects finer-scale controls not captured at this resolution. Together, these patterns provide a quantitative basis for interpreting how regional geology, climate and land-use history interact to shape soil fertility gradients across Mexico.

3.3.3. Analysis of soil properties across Mexico's agricultural and pasture systems

This research analyzed soil properties across 51.82 million hectares of agricultural and pasture land in Mexico, comprising 32.79 million hectares of agricultural systems (rainfed, irrigated, and moisture-dependent agriculture) and 19.03 million hectares of pasture systems

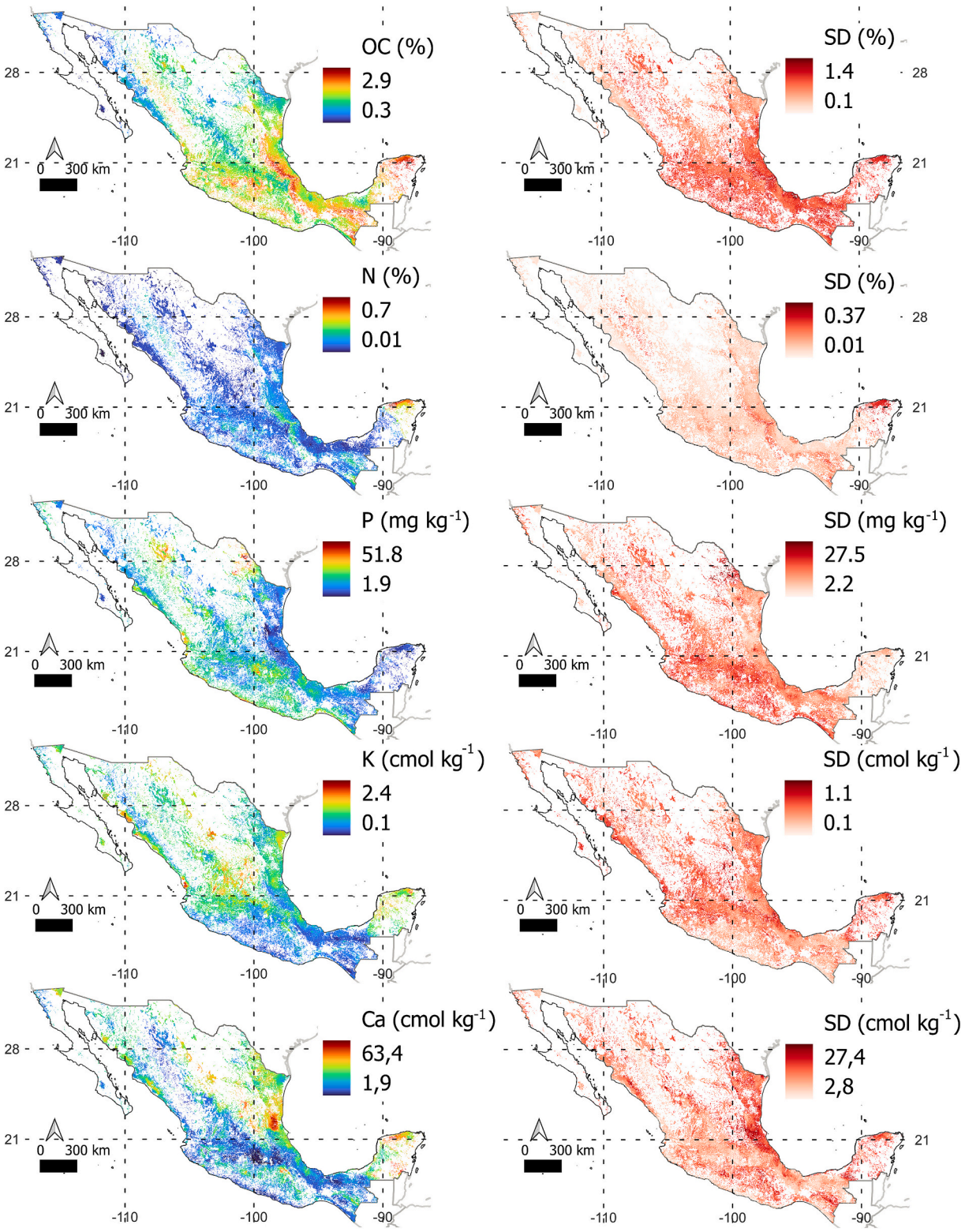


Fig. 4. Predicted mean and standard deviation of soil organic carbon (OC) and total nitrogen (N) contents, and available phosphorus (P) and exchangeable potassium (K) and calcium (Ca) concentrations across the economic regions of Mexico. Maps include coordinate grid, north arrow, and scale bar for spatial orientation (CRS: WGS84 / EPSG:4326).

(cultivated and induced pastures). Sample sizes differed among land-use systems, with rainfed agriculture and induced pastures being more densely represented in the dataset. All systems, however, included sufficient observations to support robust statistical comparisons (Table 7). ANOVA results showed highly significant effects of land use type on all analyzed soil properties ($p < 0.001$). Post-hoc comparisons confirmed statistically significant differences among systems for texture, pH, CEC, OC, N, P, K and Ca, while BD also differed significantly but within a relatively narrow range.

Sand content was highest in induced pastures and rainfed agriculture

Table 5
Regional soil properties in agricultural areas of Mexico: Physical, chemical, and nutrient content attributes across different economic regions. Values are mean followed by standard error. Capital letters indicate significant differences within each variable between regions (Tukey-adjusted, $p < 0.05$).

Region	Area (ha)	Physical and Chemical Properties						Nutrient Content				
		Sand	Silt	Clay	BD	pH (H_2O)	CEC	OC	N	P	K	Ca
		- - - - - - % - - - - - -			g cm^{-3}		$cmol_c$ kg^{-1}	- - - % - - -		mg kg^{-1}	- - $cmol_c$ kg^{-1} - -	
Central West	7,361,123	43.153 ± 5.141	24.06 ± 2.617	33.238 ± 5.569	1.297 ± 0.075	6.427 ± 0.621	23.806 ± 4.761	1.705 ± 0.275	0.152 ± 0.033	17.18 ± 3.899	0.922 ± 0.242	13.76 ± 5.198
Central South	5,392,227	44.074 ± 6.558	24.242 ± 3.179	31.44 ±5.913	1.267 ± 0.102	6.691 ± 0.794	21.102 ± 6.812	1.805 ± 0.361	0.163 ± 0.055	17.435 ± 5.796	0.759 ± 0.208	14.956 ± 8.005
Gulf of Mexico	7,084,659	35.523 ± 6.836	28.365 ± 2.986	36.685 ± 6.764	1.247 ± 0.116	6.317 ± 0.889	27.512 ± 5.082	1.935 ± 0.324	0.16 ±0.066	12.739 ± 4.215	0.657 ± 0.23	18.609 ± 9.165
Northeast	5,782,191	35.581 ± 6.302	27.848 ± 2.764	37.659 ± 6.384	1.322 ± 0.049	8.19 ± 0.369	22.326 ± 5.499	1.548 ± 0.335	0.144 ± 0.037	14.68 ± 5.852	0.968 ± 0.201	32.691 ± 5.926
Northwest	6,133,838	48.773 ± 6.508	24.486 ± 3.530	26.783 ± 5.888	1.415 ± 0.117	7.31 ± 0.848	20.955 ± 4.574	1.121 ± 0.464	0.11 ± 0.04	17.224 ± 4.422	0.884 ± 0.224	19.648 ± 6.132
North	9,537,519	45.525 ± 7.754	25.613 ± 4.393	28.973 ± 6.119	1.345 ± 0.076	7.212 ± 1.032	21.321 ± 5.344	1.479 ± 0.346	0.138 ± 0.052	20.076 ± 6.715	1.101 ± 0.242	22.758 ± 10.723
South Pacific	8,244,738	46.511 ± 6.595	24.523 ± 4.666	29.955 ± 5.169	1.286 ± 0.103	6.371 ± 0.604	23.631 ± 5.163	1.899 ± 0.261	0.166 ± 0.051	17.539 ± 5.368	0.509 ± 0.113	15.907 ± 6.021
Yucatan Peninsula	2,280,779	32.138 ± 4.552	17.542 ± 2.912	49.253 ± 7.582	1.188 ± 0.091	7.219 ± 0.408	37.028 ± 5.303	2.133 ± 0.247	0.221 ± 0.144	7.992 ± 1.904	1.223 ± 0.218	35.413 ± 8.496
National	51,817,077	43.087 ± 8.266	24.601 ± 4.448	32.548 ± 8.479	1.309 ± 0.109	6.957 ± 0.943	23.690 ± 6.906	1.652 ± 0.438	0.152 ± 0.068	16.799 ± 6.253	0.900 ± 0.307	20.831 ± 10.45

Sample sizes (n): Central West = 5903; Central South = 4950; Gulf of Mexico = 2533; Northeast = 3243; Northwest = 5187; North = 11,208; South Pacific = 6074; Yucatan Peninsula = 3596. Sample sizes are consistent across variables (sand, silt, clay, BD, pH, CEC, OC, TN, P, K).

(~46.0% and 42.9%, respectively) and lowest in moisture-dependent agriculture (~35.0%). In contrast, clay content was greatest in moisture-dependent agriculture and cultivated pastures (~39–37%), and lowest in induced pastures (~29.6%). Silt content showed smaller but significant contrasts, with the highest values in moisture-dependent agriculture. BD ranged from approximately 1.28 to 1.35 g cm^{-3} across systems, with the lowest values observed in cultivated pastures and moisture-dependent agriculture, and the highest in irrigated agriculture. Although statistically significant, these differences indicate limited variability in soil physical condition at the national scale.

Soil pH varied markedly among systems (Table 7), with irrigated agriculture exhibiting the most alkaline conditions (mean pH ~ 7.55), while moisture-dependent agriculture showed the most acidic values (≈6.44). CEC was highest in cultivated pastures and moisture-dependent agriculture (~26.6–26.7 $cmol_c$ kg^{-1}), and lowest in induced pastures (~21.8 $cmol_c$ kg^{-1}).

OC and N contents were significantly higher in pasture systems and moisture-dependent agriculture (Table 7), whereas irrigated agriculture exhibited the lowest OC and N values. Available P showed strong contrasts, being lowest in cultivated pastures and highest in induced pastures and irrigated systems. Exchangeable K and Ca were highest under irrigated agriculture, with elevated values also observed in cultivated pastures, reflecting greater nutrient accumulation under intensified management. Overall, these patterns indicate that land use and management exert a strong control on soil nutrient status and chemical properties, while physical attributes such as BD show a more moderate response at broad spatial scales.

4. Discussion

4.1. Spatial and agricultural variability

The results of this study emphasize the importance of understanding soil properties at regional and state levels to inform land management practices across Mexico. While many soil attributes are strongly influenced by inherent soil-forming factors, management practices can substantially modify properties such as soil OC and pH, particularly in regions characterized by intensive or long-standing agricultural use. The pronounced variability observed across the Yucatán Peninsula, Northern Mexico, and Central-South Mexico highlights the need for regionally tailored strategies to sustain soil health and agricultural productivity.

In the Yucatán Peninsula, the combination of high CEC, elevated OC and exchangeable Ca contents, moderately alkaline pH, and low P availability reflects a distinctive edaphic configuration associated with carbonate-dominated parent materials. The consistently high OC contents observed in this region are likely driven by stabilization mechanisms characteristic of carbonate-rich soils, including physical protection and organo–carbonate associations linked to secondary carbonates derived from limestone. These mechanisms may reduce microbial accessibility and slow organic matter mineralization, thereby promoting OC persistence, particularly under land-use histories characterized by long fallow periods and traditional management practices (Aryal et al., 2018). Such conditions are typical of calcareous karstic environments, where carbonate-rich substrates buffer acidity but favor processes that reduce P availability through precipitation and sorption reactions (Santillán et al., 2021; Campo and Merino, 2019). This pattern is consistent with the national dataset compiled by Arroyo-Cruz et al. (2025), which showed that a large proportion of Mexican agricultural soils exhibit alkaline conditions that constrain P and micronutrient availability. In karstic landscapes such as the Yucatán Peninsula, this pervasive alkalinity is therefore likely to further limit P use efficiency, despite relatively high OC contents and base status.

Agricultural expansion in the region is further constrained by the dominance of shallow Leptosols, characterized by limited soil depth, low water retention capacity, and frequent rocky outcrops (Bautista et al., 2011). While such conditions may enhance root aeration, they strongly restrict water and nutrient storage, complicating crop and pasture management. Although tendencies toward lower BD and N contents were observed, these differences were not statistically significant. Increasing drought frequency associated with declining precipitation further exacerbates these limitations (Cuervo-Robayo et al., 2020), disrupting nutrient cycling and reducing plant nutrient uptake (Campo and Merino, 2019). Deforestation and overgrazing remain major drivers of soil degradation in the region (Figueroa et al., 2020; Ellis et al., 2017), with overgrazing contributing to compaction and nutrient depletion (Enríquez Quiroz et al., 2021). Sustainable practices such as rotational grazing, agroforestry, and organic amendments are therefore essential to maintain soil structure and long-term productivity. Overall, the Yucatán Peninsula illustrates the paradox of relatively high soil fertility potential constrained by shallow soil depth and structural fragility, demanding

Table 6
State-level soil properties in agricultural areas of Mexico: Physical, chemical, and nutrient content attributes. Values are mean followed by standard error.

State	Area (ha)	Physical and Chemical Properties						Nutrient Content				
		Sand	Silt	Clay	BD	pH (H_2O)	CEC	OC	N	P	K	Ca
		- – – – – % - – – – –			g cm^{-3}		cmo_c kg^{-1}	- – - % - - -		mg kg^{-1}	- – $cmol_c$ kg^{-1} - -	
Aguascalientes	276,156	49.028 ± 2.706	23.355 ± 1.300	26.166 ± 2.628	1.338 ± 0.034	6.767 ± 0.401	21.626 ± 2.886	1.332 ± 0.172	0.121 ± 0.019	15.481 ± 1.749	1.277 ± 0.191	16.139 ± 3.345
Baja California	556,207	53.808 ± 6.185	23.841 ± 3.179	20.738 ± 5.012	1.517 ± 0.072	7.583 ± 0.445	18.901 ± 2.967	0.85 ± 0.207	0.087 ± 0.014	18.034 ± 3.246	0.800 ± 0.138	16.887 ± 5.286
Baja California Sur	215,860	48.913 ± 4.273	24.371 ± 2.316	28.02 ± 2.842	1.457 ± 0.075	7.985 ± 0.291	19.8 ± 3.971	0.619 ± 0.191	0.059 ± 0.017	15.076 ± 4.538	1.044 ± 0.117	20.810 ± 2.960
Campeche	973,383	31.007 ± 4.246	17.14 ± 2.779	50.315 ± 6.663	1.248 ± 0.034	7.044 ± 0.318	34.222 ± 3.934	1.972 ± 0.178	0.127 ± 0.020	8.522 ± 1.746	1.194 ± 0.251	30.577 ± 7.513
Chiapas	3,347,526	42.497 ± 5.871	27.077 ± 5.267	31.257 ± 5.581	1.219 ± 0.111	6.027 ± 0.411	24.76 ± 5.02	2.021 ± 0.301	0.183 ± 0.059	13.554 ± 4.749	0.462 ± 0.113	14.573 ± 5.526
Chihuahua	2,776,508	49.203 ± 4.336	23.288 ± 2.824	26.544 ± 3.233	1.359 ± 0.075	6.723 ± 0.943	19.409 ± 4.049	1.473 ± 0.361	0.153 ± 0.063	21.983 ± 5.809	1.067 ± 0.203	18.108 ± 9.174
Ciudad de México	43,640	45.856 ± 3.733	24.641 ± 2.26	27.307 ± 5.833	1.186 ± 0.101	6.153 ± 0.663	17.507 ± 6.578	1.959 ± 0.249	0.182 ± 0.040	22.194 ± 2.794	0.676 ± 0.186	7.965 ± 4.223
Coahuila de Zaragoza	1,149,945	38.932 ± 4.672	30.241 ± 1.795	32.758 ± 3.689	1.358 ± 0.046	8.377 ± 0.215	17.507 ± 3.007	1.319 ± 0.293	0.118 ± 0.025	26.361 ± 8.674	0.947 ± 0.192	28.834 ± 6.190
Colima	249,104	47.887 ± 8.513	23.44 ± 4.383	27.686 ± 5.42	1.310 ± 0.220	6.516 ± 1.202	18.339 ± 4.278	1.655 ± 0.305	0.131 ± 0.035	21.439 ± 5.39	0.634 ± 0.141	14.323 ± 5.641
Durango	1,595,852	48.232 ± 5.565	23.576 ± 3.364	26.946 ± 3.874	1.349 ± 0.087	6.769 ± 0.991	22.033 ± 5.622	1.500 ± 0.371	0.146 ± 0.051	19.39 ± 5.407	1.068 ± 0.187	19.756 ±10.651
Guanajuato	1,712,453	40.947 ± 5.119	22.866 ± 1.732	35.852 ± 6.309	1.307 ± 0.028	6.809 ± 0.581	24.224 ± 4.858	1.518 ± 0.207	0.129 ± 0.018	15.345 ± 2.153	1.022 ± 0.202	12.52 ± 5.020
Guerrero	1,916,891	49.59 ± 4.541	22.946 ± 3.334	28.505 ± 4.487	1.321 ± 0.07	6.539 ± 0.491	23.339 ± 5.455	1.849 ± 0.156	0.161 ± 0.039	19.509 ± 4.94	0.495 ± 0.071	16.193 ± 6.417
Hidalgo	1,140,617	41.134 ± 7.636	26.013 ± 2.437	33.458 ± 6.394	1.258 ± 0.092	6.939 ± 0.773	21.676 ± 6.091	1.844 ± 0.385	0.176 ± 0.061	13.983 ± 4.961	0.874 ± 0.210	16.467 ± 6.812
Jalisco	2,881,134	43.229 ± 4.489	24.795 ± 2.26	33.074 ± 4.543	1.308 ± 0.05	6.312 ± 0.558	24.486 ± 4.493	1.667 ± 0.237	0.151 ± 0.028	18.849 ± 3.625	0.993 ± 0.211	14.359 ± 4.874
Michoacán de Ocampo	2,241,799	43.317 ± 4.936	23.845 ± 2.967	33.163 ± 5.571	1.274 ± 0.087	6.347 ± 0.561	23.464 ± 4.784	1.884 ± 0.233	0.169 ± 0.035	15.724 ± 3.703	0.777 ± 0.201	13.519 ± 5.676
Morelos	271,087	45.844 ± 2.665	21.565 ± 2.234	33.594 ± 3.884	1.300 ± 0.055	6.902 ± 0.472	25.494 ± 3.882	1.818 ± 0.148	0.152 ± 0.027	20.038 ± 2.65	0.58 ± 0.061	19.524 ± 5.367
México	1,325,896	45.24 ± 2.985	24.167 ± 2.591	29.78 ± 3.567	1.251 ± 0.094	6.143 ± 0.589	16.319 ± 5.557	1.845 ± 0.33	0.158 ± 0.037	21.193 ± 5.361	0.663 ± 0.147	8.876 ± 4.966
Nayarit	759,955	41.302 ± 5.059	26.394 ± 3.745	34.398 ± 4.921	1.274 ± 0.115	5.979 ± 0.611	25.699 ± 4.292	1.78 ± 0.204	0.136 ± 0.033	22.337 ± 4.254	0.967 ± 0.205	13.211 ± 3.725
Nuevo León	2,029,654	37.163 ± 4.961	29.103 ± 2.44	35.891 ± 5.013	1.323 ± 0.055	8.259 ± 0.347	20.306 ± 3.993	1.469 ± 0.3	0.138 ± 0.038	17.186 ± 5.698	0.937 ± 0.217	31.252 ± 5.369
Oaxaca	2,972,799	46.862 ± 7.217	23.92 ± 4.352	30.31 ± 5.127	1.309 ± 0.097	6.496 ± 0.719	22.866 ± 4.792	1.843 ± 0.276	0.155 ± 0.051	19.096 ± 4.297	0.565 ± 0.124	16.737 ± 5.758
Puebla	1,821,333	45.011 ± 7.331	23.479 ± 3.958	31.012 ± 6.12	1.268 ± 0.132	6.829 ± 0.865	24.051 ± 6.48	1.819 ± 0.398	0.171 ± 0.069	16.455 ± 5.636	0.711 ± 0.183	18.300 ± 7.865
Querétaro	474,171	41.331 ± 4.868	23.922 ± 2.088	35.179 ± 4.361	1.301 ± 0.039	7.066 ± 0.627	24.175 ± 5.361	1.671 ± 0.253	0.14 ± 0.029	15.235 ± 2.417	0.977 ± 0.194	18.458 ± 8.785
Quintana Roo	278,476	33.075 ± 3.818	16.999 ± 2.412	50.597 ± 6.169	1.143 ± 0.093	7.281 ± 0.569	37.868 ± 4.698	2.269 ± 0.234	0.281 ± 0.129	7.654 ± 1.092	1.126 ± 0.172	42.215 ± 6.161
San Luis Potosí	1,802,194	37.511 ± 9.061	29.672 ± 3.993	34.924 ± 9.035	1.290 ± 0.058	7.868 ± 0.501	25.914 ± 5.525	1.656 ± 0.314	0.125 ± 0.048	14.981 ± 4.227	1.225 ± 0.319	31.655 ± 9.847
Sinaloa	2,163,673	46.68 ± 6.555	24.293 ± 3.179	29.517 ± 4.784	1.356 ± 0.065	6.898 ± 0.518	23.907 ± 4.116	1.340 ± 0.372	0.124 ± 0.036	18.787 ± 2.715	0.839 ± 0.197	17.446 ± 5.165
Sonora	2,438,914	50.718 ± 4.551	24.141 ± 3.574	24.642 ± 3.424	1.452 ± 0.093	7.762 ± 0.55	18.895 ± 3.286	0.945 ± 0.362	0.108 ± 0.042	14.971 ± 3.462	0.881 ± 0.248	23.198 ± 4.997
Tabasco	1,588,974	38.564 ± 6.291	28.403 ± 4.37	33.122 ± 6.324	1.316 ± 0.048	5.827 ± 0.311	24.642 ± 2.757	1.798 ± 0.237	0.102 ± 0.022	14.487 ± 4.193	0.449 ± 0.111	13.194 ± 4.861
Tamaulipas	3,752,667	33.714 ± 7.153	26.421 ± 2.422	39.777 ± 7.162	1.321 ± 0.041	8.112 ± 0.376	24.707 ± 6.042	1.642 ± 0.349	0.15 ± 0.034	11.724 ± 4.421	1.004 ± 0.174	34.402 ± 6.052
Tlaxcala	317,460	51.359 ± 3.265	23.383 ± 1.984	23.217 ± 1.776	1.304 ± 0.067	6.548 ± 0.306	14.169 ± 2.19	1.500 ± 0.27	0.129 ± 0.025	22.999 ± 2.658	0.740 ± 0.101	7.314 ± 1.996
Veracruz de Ignacio de la Llave	5,499,669	34.746 ± 6.702	28.34 ± 2.485	37.596 ± 6.533	1.228 ± 0.121	6.441 ± 0.942	28.224 ± 5.283	1.979 ± 0.331	0.176 ± 0.065	12.295 ± 4.100	0.710 ± 0.221	19.950 ± 9.412
Yucatán	1,029,366	33.626 ± 5.125	18.958 ± 3.36	45.777 ± 9.323	1.110 ± 0.088	7.495 ± 0.207	41.711 ± 4.709	2.322 ± 0.166	0.353 ± 0.164	7.240 ± 2.396	1.357 ± 0.109	38.744 ± 6.346
Zacatecas	2,213,698	49.875 ± 4.712	24.375 ± 3.883	26.157 ± 3.279	1.372 ± 0.064	7.07 ± 0.936	22.499 ± 3.855	1.370 ± 0.206	0.125 ± 0.028	16.77 ± 2.825	1.235 ± 0.19	20.717 ± 8.655

Sample sizes (n) per state: Aguascalientes = 160; Baja California = 644; Baja California Sur = 256; Campeche = 1851; Chiapas = 1841; Chihuahua = 3756; Mexico City = 60; Coahuila = 1535; Colima = 188; Durango = 2466; Guanajuato = 1034; Guerrero = 2183; Hidalgo = 1185; Jalisco = 2529; Estado de México = 1205; Michoacán = 2012; Morelos = 247; Nayarit = 812; Nuevo León = 1713; Oaxaca = 2003; Puebla = 1510; Querétaro = 476; Quintana Roo = 825; San Luis Potosí = 2078; Sinaloa = 1004; Sonora = 2489; Tabasco = 528; Tamaulipas = 1521; Tlaxcala = 254; Veracruz = 2016; Yucatán = 931; Zacatecas = 1382.

Table 7
Soil properties across agricultural and pasture systems in Mexico: Physical, chemical, and nutrient attributes. Values are means followed by standard errors. Capital letters indicate significant differences within each variable between systems (Tukey-adjusted, $p < 0.05$); means followed by the same letter are not significantly different.

Type of agriculture and pasture	Area (ha)	Physical and Chemical Properties						Nutrient Content				
		Sand	Silt	Clay	BD	pH (H_2O)	CEC	OC	N	P	K	Ca
		- - - - - - % - - - - - -			g cm^{-3}		$cmol_c$ kg^{-1}	- - - % - - -		mg kg^{-1}	- - $cmol_c$ kg^{-1} - -	
Moisture agriculture	207,402	34.96 ± 6.09 A	26.68 ± 5.03 A	39.08 ±7.04 A	1.29 ±0.05 AB	6.46 ±0.79 A	26.77 ± 5.35 A	1.81 ±0.27 A	0.13 ± 0.03 A	15.65 ± 6.01 A	0.79 ±0.28 A	19.89 ±9.79 A
Irrigated agriculture	10,100,933	42.68 ± 8.60 B	25.19 ±4.66 B	32.99 ±8.45 B	1.35 ±0.09 C	7.56 ±0.79 B	23.63 ±6.92 B	1.35 ±0.45 B	0.12 ± 0.05 B	17.38 ± 6.69 B	1.08 ±0.31 B	24.30 ±10.31 B
Rainfed agriculture	22,477,481	42.92 ± 8.47 B	25.02 ±4.58 B	32.28 ±8.45 C	1.30 ±0.11 A	6.89 ±0.94 C	23.69 ±6.71 B	1.69 ±0.40 C	0.16 ± 0.07 C	16.72 ± 5.94 A	0.91 ±0.32 C	20.05 ±10.39 A
Cultivated pasture	12,998,641	39.74 ± 8.76 C	23.33 ±5.19 C	37.27 ±10.25 D	1.28 ±0.13 B	7.09 ± 0.95 D	26.64 ±8.59 A	1.77 ±0.48 A	0.17 ±0.09 D	13.99 ± 7.26 C	0.83 ±0.30 A	26.07 ±12.09 C
Induced pasture	6,032,619	46.04 ± 6.12 D	24.25 ±3.39 D	29.59 ±5.59 E	1.32 ±0.11 D	6.77 ±0.89 E	21.80 ±5.45 C	1.61 ±0.41 D	0.15 ±0.05 C	18.58 ± 5.28 D	0.85 ±0.25 D	17.75 ±7.66 D

Sample sizes (n) per type of agriculture and pasture: Moisture-dependent agriculture = 254; Irrigated agriculture = 5040; Rainfed agriculture = 19,179; Cultivated pasture = 6839; Induced pasture = 11,382. Sample sizes are consistent across all variables (sand, silt, clay, BD, pH, CEC, OC, TN, P, K, Ca).

management approaches specifically adapted to karstic systems.

In Northern Mexico, soils exhibit pronounced structural and functional limitations, most notably low OC contents, reflecting arid to semi-arid climatic conditions combined with long-term management pressures. Limited biomass inputs, high evapotranspiration, and restricted organic matter turnover constrain the accumulation and stabilization of soil OC, resulting in coarse-textured soils with low buffering capacity and reduced resilience to disturbance. These intrinsic constraints are commonly reinforced by intensive grazing, mechanization, and monocropping, which reduce porosity, impair infiltration, and restrict root development, thereby limiting biological activity and nutrient cycling (Usman et al., 2018; Ahmed et al., 2019). Overgrazing accelerates erosion and desertification processes (Manzano et al., 2000), while mechanized agriculture promotes structural degradation and organic matter losses (Granados-Sánchez et al., 2013; Santini et al., 2022).

Irrigation-dependent production systems further intensify pressure on soil and water resources, particularly in highly productive regions such as Sonora, where long-term irrigation has contributed to groundwater depletion and soil salinization (Lee et al., 2021; Corwin, 2021). Although regional-scale differences in total soil N are not pronounced, the combination of low organic OC contents and relatively high levels of available P in several northern regions points to a strong reliance on external nutrient inputs rather than on internally regulated soil fertility. This dependency is reinforced by Mexico's structural reliance on imported mineral fertilizers, particularly N and P (Dobermann, 2005; Cárcamo Mallen et al., 2023; SADER, 2025). Despite widespread fertilizer application—especially in maize-based systems—nutrient use efficiency remains low and yield gaps persist (INECC–SEMARNAT, 2018; Saynes-Santillán et al., 2019), highlighting the limited capacity of these systems to translate nutrient inputs into stable soil fertility gains. Northern Mexico therefore exemplifies agroecosystems where climatic stress, coarser soil texture, and management intensity converge to produce highly productive yet structurally vulnerable systems, underscoring the urgency of integrated soil and water management strategies to enhance long-term sustainability.

In contrast to northern regions, where soil fertility is strongly shaped by climatic stress and external inputs, Central-South Mexico illustrates a different limitation pathway. Under predominantly rainfed conditions, soil fertility constraints in this region arise from limitations in nutrient retention, use efficiency, and system stability, rather than from severe nutrient depletion. Slightly acidic pH conditions observed in states such as Mexico City, Tlaxcala, and the State of Mexico reflect both intrinsic soil-forming processes associated with more weathered parent materials and long-term nutrient removal under rainfed management. These patterns are consistent with the dominance of weathered volcanic soils such as Andosols, Phaeozems, and Cambisols in the region, which are commonly characterized by acidic pH, elevated Al contents, and limited buffering capacity (Sotelo-Ruiz et al., 2021). At the regional scale, these conditions are associated with moderate to low CEC and relatively low Ca contents, indicating reduced chemical resilience and increased susceptibility to nutrient losses. Although average available P contents are intermediate to relatively high compared with other regions, soil acidity likely reduces P use efficiency through enhanced sorption to Fe and Al oxides and through surface losses associated with erosion (Sotelo-Ruiz et al., 2021). Under these conditions, nutrient limitations are therefore mediated primarily by soil chemical constraints and physical export processes rather than by low total nutrient concentrations alone. This contrasts with carbonate-rich systems such as the Yucatán Peninsula, where high base status coexists with low P availability, and with northern regions, where nutrient availability is more strongly shaped by irrigation and external inputs.

In this context, the combination of relatively coarse textures, sloping terrain, and seasonal rainfall points to erosion as an important degradation pathway in Central-South Mexico, particularly in states such as Guanajuato and Querétaro, where monocropping systems, rainfed agriculture, and limited soil conservation practices prevail (Granados-Sánchez et al., 2013). These conditions favor surface runoff and the progressive loss of soil OC and particulate nutrients, even in the absence of extreme bulk density values. This interpretation is consistent with the Sub-index of Sustainable Land Use proposed by Arroyo-Cruz et al. (2025), which classified a large proportion of rainfed agricultural soils in this region as marginal or sensitive. In contrast, cultivated pastures in Central-South Mexico exhibit higher OC, N, and CEC contents than rainfed croplands, highlighting the strong influence of land use and management on soil functioning. Practices that promote continuous ground cover, organic inputs, and controlled grazing appear to partially offset inherent edaphic constraints associated with acidity and weathering. Conservation-oriented strategies such as terracing, agroforestry, and integrated crop–livestock systems are therefore critical to reducing erosion, enhancing nutrient retention, and stabilizing soil fertility under rainfed conditions (Delgado et al., 2010). Taken together, these regional patterns highlight that soil fertility constraints across Mexico are not uniform, but arise from distinct combinations of parent material, climate, and land-use history. While the Yucatán Peninsula is

characterized by chemically buffered but physically shallow systems, Northern Mexico concentrates productive yet structurally vulnerable agroecosystems dependent on irrigation and external inputs, and Central-South Mexico represents a transitional zone where gradual degradation is driven by erosion, acidification, and management intensity. This regional differentiation underscores the value of spatially explicit soil information for diagnosing dominant degradation pathways and for designing management strategies and soil policies that are both region-specific and process-oriented, rather than relying on uniform national prescriptions.

4.2. Mexican actions for soils

The regional and system-level soil fertility constraints identified in this study align closely with the objectives of the FAO's GSNmap, which seeks to support evidence-based soil management and policy decisions at national and subnational scales. By providing spatially explicit information on soil properties and nutrients across agricultural lands and grasslands, our results demonstrate the capacity of DSM to identify dominant fertility limitations emerging from the interaction of inherent soil properties, climate, and land-use history. Previous applications of the GSNmap framework illustrate its policy relevance. For instance, Konoplina et al. (2024) showed how DSM products can identify risks of soil organic matter depletion associated with climatic stressors and agricultural land-use intensity in western Russia. In Mexico, the open-access soil dataset compiled by Arroyo-Cruz et al. (2025), covering more than 4000 agricultural sites, is already being integrated into the FAO GSNmap initiative, reinforcing the importance of linking national-scale analyses with global soil information systems to guide soil management strategies.

Our results highlight that soil fertility constraints in Mexico are highly differentiated across regions and agricultural systems, underscoring the need for region-specific and system-specific interventions rather than uniform national prescriptions. Contrasts among the Yucatán Peninsula, Northern Mexico, and Central-South Mexico demonstrate how soil texture, pH, CEC, OC, and nutrient availability interact with land use and climate to shape agricultural vulnerability and management priorities. Similarly, differences among irrigated agriculture, rainfed systems, moisture-dependent agriculture, and pasture systems confirm that management intensity is a critical driver of soil nutrient status. In addition, beyond biophysical gradients, our findings support the use of economic regions as an operational scale for soil governance. While ecoregions capture natural variability, economic regions group states with shared production systems, land-use histories, and socioeconomic contexts, providing a pragmatic framework for translating soil information into targeted policies and management actions. For example, highly productive irrigated regions such as *El Bajío* (a major agricultural region centered in Guanajuato and Querétaro, with partial extensions into Michoacán, Jalisco, Aguascalientes, and San Luis Potosí) and irrigation-dependent drylands in Northern Mexico illustrate how soil nutrient maps that integrate both biophysical and socio-productive dimensions can inform differentiated interventions (Ortiz-García et al., 2023).

Region-specific priorities emerge clearly from this analysis. In the Yucatán Peninsula, actions should prioritize improving phosphorus use efficiency under alkaline, carbonate-rich conditions, while simultaneously addressing physical constraints associated with shallow Leptosols, high rockiness, and limited water storage capacity (Bautista et al., 2011). Preventing further deforestation and overgrazing remains essential to maintaining soil structure and organic matter stocks (Ellis et al., 2017; Figueroa et al., 2020; Enríquez Quiroz et al., 2021), particularly under projected increases in drought frequency (Cuervo-Robayo et al., 2020). In Northern Mexico, soil management strategies must address the combined pressures of aridity, mechanization, and irrigation-dependent agriculture, which increase vulnerability to compaction, erosion, groundwater depletion, and salinization (Manzano et al., 2000; Granados-Sánchez et al., 2013; Usman et al., 2018; Ahmed et al., 2019; Lee et al., 2021; Corwin, 2021). These challenges are compounded by Mexico's strong dependence on imported mineral fertilizers (Dobermann, 2005; Cárcamo Mallen et al., 2023; SADER, 2025), and by persistently low nutrient use efficiency despite widespread fertilizer application (INECC–SEMARNAT, 2018; Saynes-Santillán et al., 2019). Integrated soil–water–nutrient management approaches are therefore critical to enhancing long-term system resilience. In Central-South Mexico, where rainfed agriculture predominates, soil policies should prioritize erosion control, nutrient retention, and practices that enhance system stability. The classification of large areas as marginal or sensitive in terms of sustainable land use (Arroyo-Cruz et al., 2025) highlights the need to reduce topsoil losses and improve soil functioning under sloping and erosion-prone conditions. The comparatively higher OC, N, and CEC observed in cultivated pastures demonstrate that land-use and management choices can partially offset inherent edaphic constraints, supporting the promotion of conservation-oriented practices such as terracing, agroforestry, and integrated crop–livestock systems (Delgado et al., 2010).

Finally, these regionally differentiated constraints are embedded within Mexico's dual agricultural structure, where a small number of large-scale commercial farms coexist with a majority of smallholder systems operating under limited resources and higher degradation risk (Turrent-Fernández et al., 2017; Saynes-Santillán et al., 2019). Addressing soil degradation at scale will therefore require policies that account for both biophysical heterogeneity and socioeconomic asymmetries, improving access to information, technology, and institutional support (Glover et al., 2016; Lema et al., 2022; Holt-Giménez et al., 2021). Although Mexico has developed relevant institutions and regulatory instruments for soil management (Ortiz-García et al., 2023), greater coherence, continuity, and participatory implementation are needed to translate these frameworks into effective action (Ortiz-García et al., 2022). In this context, the spatially explicit soil maps and analyses presented here provide a robust quantitative foundation to support process-oriented, regionally differentiated soil policies, bridging scientific evidence with actionable soil governance.

4.3. Strengths and limitations

Our study provides the first national digital maps of soil properties and nutrients for Mexico generated from harmonized national soil observations and DSM methodologies within the framework of the FAO's GSNmap program. Unlike previous compilations, the maps deliver spatially continuous information at pixel level together with model-based uncertainty estimates, produced through locally calibrated modeling. This represents both a methodological and applied contribution aligned with GSNmap, with direct relevance for soil management and policy-making in Mexico. Importantly, this approach helps bridge a critical information gap by connecting local management needs with the global challenge of nutrient imbalance (Richardson et al., 2023).

The accuracy assessment indicates heterogeneous but overall robust predictive performance across soil properties and nutrients. Texture variables and soil pH were predicted with strong agreement, supporting high-confidence interpretation of national-scale gradients. Moderate-to-strong performance was obtained for CEC, OC, exchangeable K and Ca, while BD showed consistent predictions. In contrast, N and available P were the most difficult variables to predict. This pattern is consistent with the strong sensitivity of N—and particularly P—to localized management practices (e.g., fertilization regimes, cropping history, grazing intensity), heterogeneous sampling density, and geochemical controls that can vary sharply over short distances (Arrouays et al., 2014; Wei et al., 2015). Accordingly, contrasts involving pH, texture, CEC, and exchangeable bases can be interpreted with high confidence, whereas nutrient-related patterns—especially for P—should be applied more cautiously for field-scale decision-making and preferably in combination with local soil testing and uncertainty information.

These results are consistent with findings reported in other country-wide GSNmap implementations. For example, Suleymanov et al. (2024) showed that soil texture and pH can be predicted with high accuracy in France, while OC, P, and K exhibited greater uncertainty due to strong influences of management and lithological heterogeneity. Similarly, Konoplina et al. (2024) reported comparable limitations in Russia, underscoring the shared challenges of nutrient mapping at national scale across contrasting agro-environmental contexts. Such international comparisons provide methodological benchmarks and practical references for communicating uncertainty and framing realistic use cases for national soil information systems.

A key strength of this study lies in its explicit focus on agricultural and grassland soils, which are central to food security and sustainable land management in Mexico. By combining harmonized national soil data with remote sensing, terrain, and climatic covariates within the SCORPAN framework, the approach ensures broad spatial coverage and operational relevance. The use of vegetation indices and agricultural land-use classes captures key organism and management components, while DEM-derived covariates provide fine-grained relief information that enhances model performance and spatial realism (Mosleh et al., 2016; Iticha and Takele, 2019). Together, these elements enabled the generation of spatially continuous maps with pixel-level uncertainty estimates, advancing beyond previous point-based compilations and aligning with the broader transition from research-oriented DSM toward operational and policy-relevant applications (Kidd et al., 2020).

Despite these strengths, limitations remain. First, nutrient mapping at national scale is constrained by legacy sampling heterogeneity and the strong imprint of management history, particularly for available P and, to a lesser extent, N. Second, the representativeness and density of soil observations differed among variables, with fewer records available for P and BD than for most other properties, which likely contributes to higher uncertainty in these layers. Third, although climatic covariates were incorporated, explicit parent material and lithological information remains critical for improving the interpretation of base status and geochemical constraints on nutrients such as P and K; future integration of soil-forming material proxies, lithology classes, or carbonate occurrence layers would likely enhance both predictive performance and interpretability (Arrouays et al., 2014; Wei et al., 2015). Fourth, model performance was evaluated using k-fold cross-validation and an independent hold-out validation; however, accuracy estimates may still be optimistic in the presence of spatial autocorrelation, and future work should complement these approaches with spatially blocked validation to obtain more conservative estimates (Wadoux et al., 2022). Finally, expanding soil monitoring networks and improving representativeness in under-sampled regions would further strengthen model reliability and reduce uncertainty, particularly for nutrient variables. Together, these strengths and limitations reinforce the value of DSM approaches for national-scale soil property and nutrient mapping while clarifying the scope of confidence in the predictions.

5. Conclusions

This study delivers the first national digital maps of soil properties and nutrients for Mexico derived from harmonized soil observations and DSM methodologies within the framework of the FAO GSNmap. By providing spatially continuous predictions accompanied by model-based uncertainty estimates, the maps represent a substantial advance over previous point-based compilations and offer an operational foundation for soil assessment at regional and national scales. The results demonstrate that soil fertility constraints in Mexico are not governed by a single national pattern, but instead emerge from distinct combinations of climate, soil chemical and physical properties, and land-use history. These interacting factors shape nutrient availability, retention, and use efficiency in contrasting ways across regions and agricultural systems. As a result, apparent similarities in nutrient concentrations can mask fundamentally different limitation pathways, underscoring the importance of process-oriented and spatially explicit diagnostics for interpreting soil fertility at scale.

The strong spatial contrasts identified among the Yucatán Peninsula, Northern Mexico, and Central-South Mexico highlight the limitations of uniform soil management strategies. Carbonate-rich karstic systems, arid and irrigation-dependent agroecosystems, and rainfed landscapes developed on weathered materials each exhibit specific constraints and vulnerabilities that require differentiated management responses. In this context, the maps presented here provide a diagnostic framework for identifying dominant degradation pathways and for prioritizing regionally adapted interventions, rather than prescribing generalized national recommendations. Differences in predictive performance among soil properties further clarify the appropriate scope of application of the maps. High accuracy achieved for texture and soil pH supports confident interpretation of national and regional gradients, while moderate-to-high performance for CEC, OC, and exchangeable bases allows their use in strategic planning and comparative assessments. In contrast, available P and total N remain more difficult to predict due to their strong sensitivity to localized management and legacy effects. Overall, this work reinforces the value of DSM as a decision-support tool for soil governance and sustainable land management in Mexico. By linking harmonized national data with global initiatives such as GSNmap, the study contributes to bridging the gap between scientific soil information and actionable policy frameworks. Continued investment in soil monitoring, improved representation of soil-forming materials, and integration of management information will be critical for refining these products and enhancing their long-term relevance for food security, environmental sustainability, and climate-resilient agriculture.

Code and data availability

The soil profile data used in this study were compiled from national and institutional databases and are subject to data-sharing agreements that restrict public redistribution. Environmental covariates were obtained from publicly available sources, including national geospatial datasets and global climate surfaces. The scripts used for data preprocessing, modeling, and validation were developed in R and are available from the corresponding author upon reasonable request, for academic and non-commercial research purposes.

CRediT authorship contribution statement

Thalita Fernanda Abbruzzini: Writing – review & editing, Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Daniela Figueroa:** Writing – review & editing, Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Carlos Eduardo Arroyo Cruz:** Writing – review & editing, Visualization, Validation, Software, Investigation, Data curation, Conceptualization. **Viviana Marcela Varón-Ramírez:** Writing – review & editing, Visualization, Software, Investigation, Conceptualization. **Mario Antonio Guevara Santamaria:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Blanca Prado:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.geodrs.2026.e01081.

Data availability

Data will be made available on request.

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